RAG Approaches for Scientific Paper Analysis

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Abstract

Scientific papers contain information spread across text, figures and tables. Text based RAG approaches may overlook important visual evidence. This project investigates how the different types of evidence (text, figures and tables) impact scientific question answering. It also develops an automated pipeline for scientific paper analysis.

The project comprises two complementary studies. The first study evaluated multimodal RAG using the arXiv subset of Open RAGBench. The dataset contains 1,000 scientific papers and 3,045 questions. Four evidence configurations were compared on 982 image associated queries: text only, text with distractors, text with figure related context and text with both figure context and image pixels. Adding figure related context resulted in a 0.01243 increase in mean semantic similarity over the distractor condition. This improvement was statistically significant but remained below the practical effect threshold of 0.02. Adding image pixels produced a further increase of only 0.00123. This difference was not statistically significant. The study had sufficient power to detect a difference of 0.02. A descriptive analysis of 14 queries suggested that pixels could help when visual interpretation was required and the correct image was retrieved.

The second study developed an automated PDF review pipeline using BGE-M3, a BGE reranker and Qwen2.5-VL. In a single paper experiment, retrieval Hit@1 increased from 0.30 with MiniLM to 0.60 with BGE-M3 and reranking. A manual audit of four documents found claim support rates between 20% and 76% across generation modes where a rate could be computed. No fabricated scientific findings were identified in the audited review outputs. Observed errors involved retrieval, evidence attribution, numerical extraction and scientific interpretation.

The results indicate that figure related text captures much of the average multimodal benefit. Image pixels appear to provide additional value for visually dependent questions when relevant images are retrieved. The PDF review findings show that structural validation alone cannot establish semantic correctness.

Keywords: *Retrieval Augmented Generation, Multimodal RAG, Scientific Document Analysis, Visual Question Answering, Evidence Grounding, Large Language Models, Vision Language Models*

1. Introduction

Scientific papers present information through text, figures and tables. Key findings often appear in plots. Numerical comparisons are usually reported in tables. Text dependent systems may have difficulty extracting and integrating this information.

Large Language Models (LLMs) can support scientific question answering and document analysis. However, their responses may include incorrect or unsupported information [1]. Retrieval Augmented Generation (RAG) aims at addressing this problem by providing the model with external information during the generation process [2]. Conventional text based RAG divides documents into smaller passages and retrieves relevant passages using embedding models. These passages are then supplied to the language model.

Figures in scientific papers are usually accompanied by a caption and a discussion in the text. When a RAG system retrieves a figure, much of its content may therefore already be available as text. Vision Language Models (VLMs) such as Qwen2.5-VL [3] can also process the image itself. It remains unclear whether this provides additional information. Some answers may be available only in the plot, such as a threshold read from an axis. In such cases, image pixels may be useful. This benefit depends on the retriever returning the correct figure. An unrelated image adds input without adding relevant evidence.

Reviewing a paper is a different task from answering a question about it. A reviewer has to find what the authors claim, how they tested it and where their evidence stops. An automated reviewer must also show where each statement comes from. This is where errors become hard to see. A system can cite the correct page and still misread it. It can report an AUC as an accuracy. It can present a result from a cited paper as the authors' own. Such outputs may still pass the structural checks. Structural validity and semantic correctness therefore have to be evaluated separately.

The pixel question is tested on Open RAGBench [4]. Each of the 982 image associated questions is answered under four evidence conditions: text alone, text with distractors, text with figure context and the same context with the figure image added. Comparing the answers shows what the image contributes once its surrounding text is already in the prompt. The review problem is addressed with a pipeline that runs entirely on local hardware. It reads a single PDF and produces a structured review in which every statement must cite a page. Its output for four papers is then checked manually against the original documents, one claim at a time.

1.1. Research Questions

Four questions follow from the problems above. The first two separate what each layer of figure evidence adds. The third asks when an image is actually needed. The fourth concerns the review pipeline.

RQ1: Does figure related context improve answers because of what it says, or would any additional text help as much?

RQ2: Does adding image pixels improve answer quality beyond the information already available in figure related textual context?

RQ3: How do visual requirements and image retrieval quality affect multimodal answer generation?

RQ4: How reliably can a scientific PDF review pipeline produce findings traced to the source while keeping structural validity apart from semantic correctness?

1.2. Main Contributions

The main contributions of this project are:

1. **Controlled multimodal RAG evaluation:** Four evidence configurations are compared across 982 image associated questions from Open RAGBench. The analysis includes significance testing and power analysis.
2. **Visual necessity and oracle image analysis:** Fourteen queries are labeled manually according to whether the image is required, helpful or not needed to answer them. In the oracle experiment, the retrieved images are then replaced with manually selected correct ones. Comparing the answers shows what incomplete or incorrect retrieval changes.
3. **Automated scientific PDF review pipeline:** A fully local system takes a single PDF and splits it into sections and smaller passages. BGE-M3 and a reranker select the most relevant passages. A separate module selects relevant page images. Qwen2.5-VL then extracts structured findings for five fixed review topics. Every statement in the final review must cite the page it came from. A

structural audit checks these citations. When the evidence is insufficient, the system says so instead of producing an answer.

4. **Manual validation and error analysis:** Each claim the pipeline generates for four papers is checked manually against the page it cites. When a claim is wrong, the audit tracks down where the error started. Many errors involve a correct value that is cited to the wrong page or credited to the wrong source. Others come from misreading, such as a minus sign lost during extraction. These findings illustrate why evidence attribution and scientific interpretation require separate verification.

2. Related Work

2.1. Retrieval Augmented Generation for Scientific Documents

Retrieval Augmented Generation (RAG) combines information retrieval with language generation. Lewis et al. [2] introduced a framework that retrieves external passages to support answer generation. The quality of these passages plays an important role in the resulting answers.

Dense retrieval compares queries and documents as embeddings. Compact models such as MiniLM make this practical for semantic search [5]. BGE-M3 goes further and combines dense, sparse and multi-vector retrieval in a single model [6]. Retrieved candidates can then be scored again by a BGE reranker. Unlike the retriever, this cross-encoder reads the query and each candidate together [7]. The Qwen3 Embedding series covers both steps with models for embedding and reranking [8].

In a scientific paper, the evidence for one answer may be split across several sections or between the text and a figure. Open RAGBench [4] reflects this. Its questions differ in whether the evidence comes from text alone or also from figures and tables.

2.2. Multimodal RAG and Visual Evidence Retrieval

Text based retrieval may overlook information contained in figures and document layouts. Multimodal approaches aim to address this limitation by using visual representations.

ColPali [9] treats each document page as an image instead of extracting its text. It splits the page image into small patches and compares them with the terms of the query. The best match for each term is added up to score the page. VisRAG [10] keeps the pages as images all the way to the answer. The retrieved page images go straight to a vision language model, which writes the answer from them.

Qwen2.5-VL [3], the model used in this project, has also been tested on charts from scientific papers. Schleid et al. [11] fine tuned its 7B and 32B versions for the SciVQA shared task, in which questions are asked about such charts. The larger model outperformed a GPT-4o-mini baseline and finished fourth.

MRAG-Bench evaluates vision centric retrieval scenarios in which image evidence can offer advantages over text [12]. MMDocIR distinguishes page level retrieval from the retrieval of individual document elements such as figures and tables [13]. These benchmarks emphasize that finding relevant visual content and using it to answer a question are related but distinct challenges.

The present study complements this literature by comparing figure related textual context with the same context accompanied by image pixels. Its additional visual necessity and oracle analyses examine when those pixels may be useful and whether relevant figures were actually selected. These analyses are descriptive rather than a new large scale visual retrieval benchmark.

2.3. Automated Scientific Paper Analysis and Evaluation

Large Language Models have also been investigated for automated scientific paper reviewing. Zhou et al. [14] evaluated language models on review generation and score prediction. They found that the models made frequent mistakes, struggled with long papers and rarely provided critical feedback comparable to human reviewers.

Reliable review generation also requires accurate evidence attribution. Gao et al. [15] introduced ALCE, a benchmark for evaluating generated answers with citations. Their experiments showed that existing systems frequently produced claims without complete citation support. A valid citation format therefore does not guarantee that the cited source supports the corresponding statement.

RAGAs evaluates multiple aspects of RAG quality, including the relevance of retrieved information and the faithfulness of generated responses [16]. ARES similarly separates context relevance, answer relevance and answer faithfulness [17]. For scientific literature analysis, OpenScholar investigates retrieval supported scientific synthesis [18]. Together, this literature motivates evaluating retrieval, answer quality and evidential support separately. The present project focuses on concrete errors in scientific PDF extraction, citation attribution and review generation rather than treating structurally valid references as sufficient proof of semantic correctness.

3. Materials and Methods

The project consists of two complementary components. The first evaluates the contribution of textual and visual evidence to scientific question answering. The second develops an automated system for analyzing individual scientific PDF documents. Both components use retrieval and language generation but differ in their datasets, retrieval configurations and evaluation procedures.

3.1. Dataset and Benchmark

The multimodal RAG experiments use the arXiv subset of Open RAGBench [4]. The dataset contains 1,000 scientific papers divided into 18,840 sections. It provides 3,045 questions, each with a reference answer and a link to the relevant document section.

The questions are grouped according to their evidence sources. Table 1 shows their distribution.

Table 1. Question distribution in the Open RAGBench arXiv subset

Query source	Questions
Text	1,914
Text and image	763
Text and table	148
Text, table and image	220
Total	**3,045**

The controlled multimodal evaluation focuses on questions associated with image sources. The dataset contains 983 such questions. One question was excluded because its reference answer was incomplete. The final paired evaluation contains 982 questions and 3,928 generated answers across four conditions.

3.2. Text Retrieval and Reranking

The benchmark retrieval system divides scientific paper sections into text chunks of 450 tokens with an overlap of 50 tokens.

The text retrieval branch uses all-MiniLM-L6-v2 [5], a model from the Sentence Transformers ecosystem [19], for dense retrieval. The top 20 candidates are reranked using the ms-marco-MiniLM-L-6-v2 cross encoder [5]. At most one chunk is retained per document section to reduce redundant evidence. The three highest ranked non-image chunks are selected as the final textual evidence.

The automated PDF review pipeline uses a separate retrieval configuration. Open RAGBench provides each paper already divided into sections. A raw PDF comes without this structure. The pipeline therefore detects section headings itself and segments the text at these points. Each section is then split into chunks of 240 tokens with an overlap of 40 tokens. These chunks are smaller than the 450 token chunks used for the benchmark.

BGE-M3 [6] is used for dense retrieval. A BGE cross encoder reranker [7] then reorders the retrieved candidates. This configuration is evaluated separately from the benchmark retrieval system.

3.3. Figure and Table Evidence Retrieval

In the benchmark corpus, a figure appears in the section text as a Markdown reference to an image file, such as img-4.jpeg. A textual index cannot directly represent the image pixels. Each figure is therefore represented by the passage surrounding its reference. Table chunks combine surrounding text with table content extracted in Markdown. Both evidence types are embedded as text. This makes figures and tables searchable through the words that describe them.

Table 2 summarizes the indexed evidence.

Table 2. Composition of the enriched retrieval index

Evidence type	Chunks
Text	73,294
Figure context	7,752
Table context	3,486
Total	**84,532**

The figure retrieval branch uses Qwen3 Embedding [8] to rank figure contexts. The two highest ranked figures are combined with three non-image chunks from the text retrieval branch. Image pixels are supplied separately to the Vision Language Model (VLM) in the T+C+P condition.

The controlled pixel ablation focuses on figures and does not introduce a separate table image condition.

3.4. Multimodal Answer Generation

Qwen2.5-VL-7B-Instruct [3] is used for multimodal answer generation. The model receives the question and selected evidence. Its input configuration depends on the experimental condition.

Figure 1 illustrates how the textual and visual evidence branches are combined before answer generation. Image pixels are provided only when the experimental condition explicitly includes them.

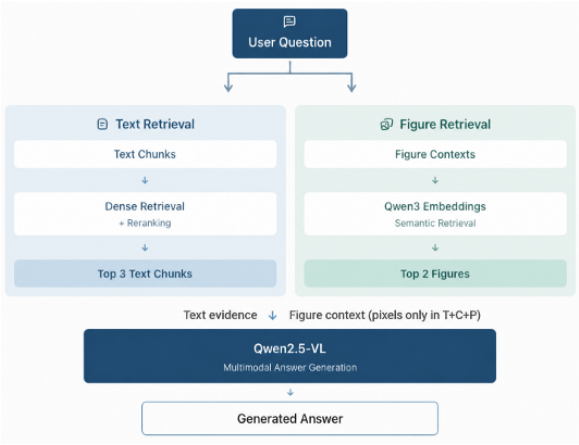


Figure 1. Architecture of the multimodal scientific question answering system. Text and figure contexts are retrieved through separate branches. Qwen2.5-VL generates answers using the selected evidence. Image pixels are supplied only in the T+C+P condition.

All four controlled ablation conditions use greedy decoding with sampling disabled (do_sample=False). The maximum generation length is 256 new tokens.

The experiments use A100 and V100 GPUs. The final T and T+D generation jobs were executed on a V100 while T+C and T+C+P were generated on an A100. A repeatability check reproduced all 30 tested answers for both T and T+D within the same V100 configuration.

This check establishes repeatability within that configuration. It does not establish identical generation across A100 and V100 hardware.

3.5. Controlled Ablation Design

Four evidence configurations are compared to examine the contribution of additional textual and visual information.

Table 3. Evidence configurations used in the controlled ablation

Condition	Evidence
T	Textual evidence only
T+D	Textual evidence with in-domain distractor context
T+C	Textual evidence with figure related textual context
T+C+P	Textual evidence with figure context and image pixels

The distractor condition controls for the effect of adding additional textual evidence. Distractors originate from the same dataset. Their assignments were checked through an audit that combined earlier screening results with manual review. All 982 evaluated questions received a final leakage label of NONE.

In T+C, the model receives figure related textual context without image pixels. T+C+P provides identical textual evidence and additionally includes the corresponding image pixels. The two conditions differ only in the inclusion of image pixels. This design isolates the effect of providing visual input while keeping textual evidence and generation settings fixed.

The primary comparisons are T+C against T+D and T+C+P against T+C. The first evaluates the contribution of figure related textual context relative to distractor context. The second isolates the additional contribution of image pixels.

3.5.1. Visual Necessity Analysis

A manually reviewed subset of 14 image associated questions is used for a descriptive analysis.

Each question receives one of three visual necessity labels:

- **REQUIRED:** Visual inspection is necessary to recover the requested information.
- **HELPFUL:** Images provide useful information beyond the available text.
- **NOT_NEEDED:** Textual evidence is sufficient to answer the question.

Answer generation is compared with and without image pixels. The analysis examines whether the effect differs across these categories.

The subset was manually selected and is not treated as a representative random sample of the complete benchmark.

3.5.2. Oracle Image Analysis

An additional experiment examines the effect of image retrieval quality.

Retrieved images are compared with manually selected reference images. Retrieval status is classified as COMPLETE, PARTIAL or MISS according to whether the relevant images are recovered.

Answers generated using retrieved images are compared with answers generated using oracle images. The textual evidence remains unchanged.

This comparison examines the effect of correcting image selection within the evaluated configuration. The analysis is descriptive because only a small number of questions have different retrieved and oracle image inputs.

3.6. Automated PDF Review Pipeline

The second component implements a fully local pipeline for scientific PDF analysis. The system combines document processing, retrieval, multimodal information extraction and evidence grounded review generation.

Figure 2 illustrates the complete workflow.

3.6.1. Document Processing and Evidence Retrieval

The pipeline begins with PDF text extraction and semantic segmentation. Section headings are detected and the extracted text is divided into overlapping chunks.

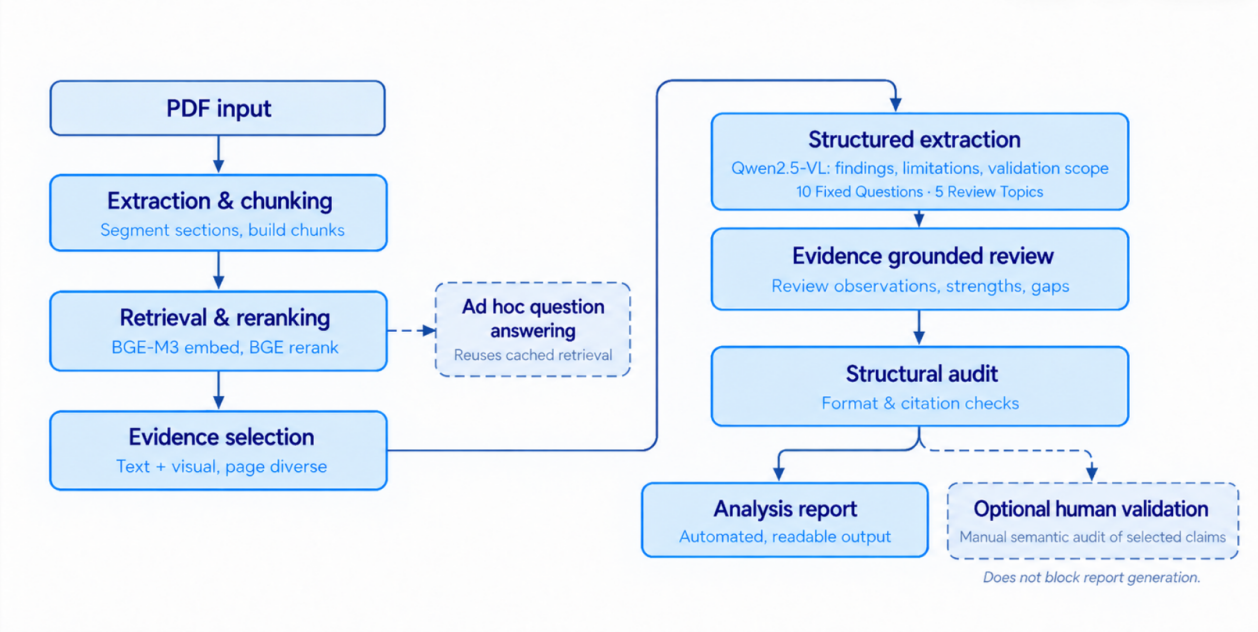


Figure 2. Workflow of the automated scientific PDF review pipeline. The system extracts document evidence, retrieves and reranks relevant passages, selects textual and visual evidence and generates structured findings. A structural audit checks the outputs before report generation. Optional manual validation supports semantic verification.

BGE-M3 retrieves relevant passages and a BGE reranker reorders the candidates. A page diversity constraint retains at most two chunks from the same page by default. This reduces the concentration of overlapping evidence from a single document region.

Visual evidence is handled differently from the benchmark figure retrieval system. The PDF review pipeline uses rendered page images rather than individually extracted scientific figures. A separate visual retrieval module evaluates the relevance of candidate PDF pages to the review question.

The pipeline supports three generation modes:

1. text_retrieval_text_only: Text retrieval followed by text only generation.
2. hybrid_retrieval_text_only: Hybrid evidence retrieval with text only generation.
3. hybrid_retrieval_with_images: Hybrid evidence retrieval with selected page images provided to the VLM.

The two hybrid modes differ in whether the selected page images are supplied to the generation model.

3.6.2. Structured Information Extraction

Qwen2.5-VL performs structured scientific information extraction using retrieved evidence.

The extraction procedure covers five predefined review topics. Each topic contains two fixed questions.

Table 4. Scientific review topics and their information requirements

Review topic	Information extracted
Purpose and contributions	Research problem and claimed contributions
Methods	Proposed methods, inputs and implementation details
Experiments and validation	Datasets, baselines, evaluation metrics and validation procedures
Findings	Quantitative and qualitative results
Limitations	Reported limitations and evidence restrictions

Each topic is processed under three generation modes. Extracted findings preserve references to the supporting evidence and source PDF pages.

The configuration supports a maximum of 1536 new tokens for verbose structured extraction tasks. This is separate from the 256 token limit used in the controlled ablation and single paper feasibility experiment.

3.6.3. Evidence Grounded Review

The extracted information is used to generate an automated scientific review. The review organizes its outputs into three categories: Strengths, Recommendations and Unresolved checks.

Each review statement references the extraction records used as supporting evidence. Invalid evidence references are removed. A statement is discarded only if no valid reference remains.

The pipeline also identifies absence claims using regular expressions. If such a statement cites no extraction record labeled as a limitation, it is classified as an unresolved check rather than a verified recommendation.

This mechanism prevents unsupported absence claims from being presented as confirmed limitations of the source document.

3.6.4. Structural Audit and Report Generation

The structural audit validates generated records and their evidence references. It checks whether cited pages belong to the retrieved evidence and whether numerical values satisfy the expected format.

It also checks whether each reported numerical value appears in the extracted text of the cited page. Values that cannot be located in the extracted page text are flagged for review. A value visible only in a rendered page image may therefore be flagged even when it is correct.

A separate heuristic identifies numbers that may have been copied from citation references. These cases are flagged for manual review rather than rejected because small integers may also represent legitimate scientific values.

These checks establish structural validity but do not guarantee semantic correctness.

The pipeline generates a readable analysis report after structural validation. Manual semantic verification is optional and does not block report generation.

The system also supports additional question answering using its cached retrieval index. It permits abstention when the available evidence is insufficient.

3.6.5. Single Paper Feasibility Experiment

A separate feasibility experiment evaluates the retrieval and answer generation stages using arXiv:2410.14077v2, a scientific paper on inverter

output impedance estimation.

Ten Open RAGBench questions associated with this paper are used with their original query identifiers, reference answers and gold section labels.

The PDF text is extracted and divided into chunks of 240 tokens with an overlap of 40 tokens. BGE-M3 retrieves the top 50 candidates. A BGE reranker reorders these candidates. The three highest ranked chunks are selected as textual evidence.

For multimodal generation, the two highest ranked distinct pages represented in the reranked candidates are rendered as images. This experiment uses a simpler image selection method than the full review pipeline. Page images are derived from the text ranking rather than scored by the visual relevance module.

Qwen2.5-VL-7B-Instruct generates answers under two conditions: text only and text with the two selected page images. Both conditions receive identical textual evidence. Generation uses greedy decoding with a maximum of 256 new tokens.

Retrieval performance is evaluated using Hit@1, Hit@3 and Hit@5. A retrieval hit is counted when a retrieved chunk belongs to the gold section.

Answer quality is measured using semantic similarity, Token F1 and ROUGE-L. A separate comparison evaluates the MiniLM retrieval baseline against BGE-M3 with reranking. An OCR ablation examines whether additional text extraction improves retrieval.

The experiment contains 10 questions from one scientific paper. Its results are interpreted as a feasibility study rather than a general performance estimate.

3.6.6. Manual PDF Review Audit

The PDF review pipeline is evaluated using four scientific documents:

1. *Inverter Output Impedance Estimation in Power Networks: A Variable Direction Forgetting Recursive-Least-Square Algorithm Based Approach* by Park et al. [20].
2. *LLM Fine-Tuning: Concepts, Opportunities, and Challenges* by Wu et al. [21].
3. *A Comprehensive Benchmark of Machine and Deep Learning Across Diverse Tabular Datasets* by Shmuel et al. [22].
4. *Parkinson's Disease Classification from Gait Dynamics Using Machine Learning and Deep Learning*, an unpublished project report by Arici and Arias Vergara [23].

The inverter impedance paper is used in both the single paper feasibility experiment and the manual PDF review audit. These experiments have different evaluation objectives.

The manual audit uses a worksheet that associates extracted claims with their cited PDF pages. A single annotator compares each claim with the original source and records whether it is supported as written.

Evidence support rates are calculated using the audited extraction records for each generation mode. A record is counted as supported when its claim is labeled Supported as written. Because rates are computed over records, repeated claims may contribute more than once.

The reported rates therefore do not estimate the proportion of unique scientific propositions supported by the source document.

Unsupported statements are examined to identify their likely error sources. The audit distinguishes retrieval errors, incorrect attribution, structured extraction errors and model or metric identification errors.

The Parkinson's disease report was authored by the annotator. All four documents were evaluated using the same manual audit procedure.

The audits were performed by a single annotator and are interpreted as descriptive case studies. Documents with very few extracted claims are not assigned meaningful percentage estimates.

3.7. Evaluation Metrics and Statistical Analysis

The experiments use retrieval metrics, automatic answer similarity metrics and manual evidence verification.

Table 5. Evaluation metrics and their purposes

Metric	Purpose
Semantic similarity	Measures semantic similarity between generated and reference answers
Token F1	Measures token overlap with reference answers
ROUGE-L	Measures longest common subsequence overlap
Hit@k	Measures retrieval success among the top k candidates
Manual evidence audit	Assesses whether generated claims are supported by source evidence

3.7.1. Automatic Answer Evaluation

Semantic similarity is the primary metric for the controlled ablation. It is computed using the all-MiniLM-L6-v2 sentence embedding model [5], which follows the Sentence-BERT approach [19].

Token F1 and ROUGE-L are used as secondary metrics, with ROUGE-L following the ROUGE framework [24]. The evaluation uses the original generated answers and reference strings without modification.

3.7.2. Statistical Significance and Power

The controlled ablation uses paired comparisons because the same 982 questions are evaluated under all four conditions.

Differences between conditions are tested using two sided paired t-tests. The Holm procedure corrects for multiple comparisons across the six pairwise comparisons.

A difference of 0.02 in mean semantic similarity is used as the practical effect threshold.

The power analysis uses a two sided paired test with a planning significance level of 0.01667, target power of 0.80 and target difference of 0.02.

The paired t-test was selected after the experimental results were available because the original protocol did not explicitly specify the statistical test in advance. This is treated as a limitation of the inferential analysis.

3.7.3. Evaluator Calibration

A separate calibration analysis examines the relationship between semantic similarity and human judgments.

Thirty generated answers are manually labeled as CORRECT, PARTIAL or INCORRECT. Their semantic similarity scores are compared across the three groups.

The analysis examines whether semantic similarity reflects differences in answer quality and whether it can distinguish correct from incorrect responses.

The manual evidence verification procedure for the PDF review pipeline is described in Section 3.6.6.

4. Experimental Results

4.1. Evaluator Calibration

Semantic similarity was evaluated against manual correctness judgments for 30 generated answers. The answers were classified as CORRECT (15), PARTIAL (8) or INCORRECT (7).

The mean semantic similarity scores were 0.7467 for CORRECT answers, 0.7720 for PARTIAL answers and 0.5129 for INCORRECT answers. The corresponding standard deviations were 0.1604, 0.0549 and 0.3238.

Although INCORRECT answers had a lower mean semantic similarity score, the score distributions overlapped. PARTIAL answers also had a slightly higher mean score than CORRECT answers. These findings indicate that semantic similarity does not reliably distinguish individual correct and incorrect answers.

A single similarity score may not reliably show whether an answer is correct. It was therefore used only to compare average performance across conditions. Token F1 and ROUGE-L were reported alongside it.

4.2. Controlled Multimodal Ablation

The controlled ablation evaluated four evidence configurations on 982 image associated questions from Open RAGBench. Each question was evaluated under all four conditions, resulting in 3,928 responses.

Table 6 reports the mean semantic similarity, Token F1 and ROUGE-L scores.

Table 6. Mean answer quality across the four evidence configurations

Condition	Semantic similarity	Token F1	ROUGE-L
T	0.7122	0.3131	0.2527
T+D	0.7157	0.3170	0.2572
T+C	0.7282	**0.3216**	**0.2632**
T+C+P	**0.7294**	0.3201	0.2611

The T condition achieved a mean semantic similarity of 0.7122. Adding distractor context increased the score slightly to 0.7157. Figure related textual context produced a larger improvement, reaching 0.7282 under T+C.

Adding image pixels resulted in the highest mean semantic similarity of 0.7294 under T+C+P. However, Token F1 and ROUGE-L were slightly lower than in T+C. The condition with the highest mean therefore depended on the evaluation metric. Bold values indicate the highest mean in each column and do not imply statistical significance.

4.2.1. Statistical Comparisons

Six pairwise comparisons were conducted using paired t-tests with Holm correction.

The comparison between T+C and T+D produced a mean semantic similarity difference of +0.01243. This difference remained statistically significant after Holm correction ($p = 0.00045$).

However, the improvement was smaller than the practical effect threshold of 0.02. The result therefore demonstrates a statistically detectable contribution from figure related textual context without reaching the specified practical threshold.

The comparison between T+C+P and T+C produced a mean difference of +0.00123. This difference was not statistically significant ($p = 0.706$) and remained below the practical effect threshold.

No pairwise comparison was statistically significant for Token F1 or ROUGE-L after multiple comparison correction.

The distractor condition also did not show an overall decrease in mean performance relative to T. Adding irrelevant context therefore did not consistently reduce answer quality in this evaluation.

4.2.2. Statistical Power

The power analysis examined whether the available sample size was sufficient to detect a semantic similarity difference of 0.02.

The paired differences had an estimated standard deviation of 0.10222. Using a two sided test with a planning significance level of 0.01667 and target power of 0.80, approximately 277 paired observations were required to detect the specified difference.

The evaluation contained 982 paired observations. The estimated power to detect a difference of 0.02 was approximately 0.9999. The minimum detectable difference at 80% power and the planning significance level of 0.01667 was approximately 0.01057.

These results indicate that the evaluation had sufficient statistical sensitivity to detect the specified practical effect under the power analysis assumptions. The observed pixel improvement of +0.00123 was substantially smaller.

However, this result does not establish that image pixels have no effect under other retrieval configurations, tasks or models.

4.3. Visual Necessity and Oracle Image Analysis

A manually reviewed subset of 14 questions was examined to investigate whether the contribution of image pixels depends on visual necessity.

Seven questions were classified as REQUIRED, one as HELPFUL and six as NOT_NEEDED.

Table 7 summarizes the visual necessity analysis and oracle image retrieval status.

Table 7. Descriptive visual necessity and oracle image results

Category	n	Result
Visual necessity		
REQUIRED	7	Mean semantic difference: +0.0317
HELPFUL	1	Mean semantic difference: +0.1297
NOT_NEEDED	6	Mean semantic difference: −0.0182
Oracle retrieval		
COMPLETE	11	Identical retrieved and oracle answers
PARTIAL	2	Incomplete image retrieval
MISS	1	Relevant image not recovered

The REQUIRED group showed a positive mean semantic similarity difference of +0.0317 when image pixels were provided. The single HELPFUL question also improved. In contrast, the NOT_NEEDED group showed a mean decrease of 0.0182.

These results suggest that visual information may be more useful when a question requires visual interpretation. However, the manually selected subset is too small to support population level conclusions.

4.3.1. Oracle Image Retrieval

The oracle analysis examined whether incomplete or incorrect image retrieval limited answer generation.

Image retrieval was classified as COMPLETE for 11 questions, PARTIAL for two questions and MISS for one question.

For all 11 COMPLETE cases, the retrieved image and oracle image conditions produced exactly the same answers. This provided a consistency check for cases in which the image inputs were unchanged.

The retrieved and oracle image inputs differed in three cases. Across these cases, replacing retrieved images with manually selected reference images produced mean improvements of +0.03916 in semantic similarity, +0.14538 in Token F1 and +0.14750 in ROUGE-L.

Semantic similarity improved in two cases and decreased in one. The mean improvement therefore did not represent a consistent benefit across all three corrected retrieval cases.

These findings indicate that image selection can affect multimodal answer quality. The results remain descriptive because only three questions received different retrieved and oracle image inputs.

4.3.2. Illustrative Cases

Two manually examined cases illustrate the relationship between visual necessity and retrieval quality.

Successful visual evidence case. The question asked at what generation size the novelty of two models began to decline significantly. The reference answer was 10^9, reported in Figure S7 of arXiv:2405.04912v2.

The relevant section moved from rank 8 to rank 2 after reranking. In T+C, the model could not determine the exact value from textual context alone. In T+C+P, the same textual context was accompanied by the retrieved figure pixels. The model then correctly identified the generation size as 10^9.

This case demonstrates that image pixels can provide useful information when the model cannot recover a required numerical value from the available textual context.

Failed visual evidence retrieval case. A second question asked whether male coverage differed between two approaches. The reference answer reported male coverage of 0.886 for KDE-HPD and 0.900 for the parametric approach in arXiv:2406.08366v2.

The correct section moved from rank 8 to rank 2 after reranking. However, an unrelated paper remained at rank 1. The two selected images came from two different unrelated papers, arXiv:2406.18052v3 and arXiv:2406.15522v3.

Both T+C and T+C+P produced incorrect answers stating that the context did not provide the requested data.

This case demonstrates that retrieving the correct section among the candidates is insufficient when final evidence selection provides unrelated images or omits the information needed to answer the question.

The two cases illustrate different success and failure mechanisms. They are not treated as representative estimates of benchmark performance.

4.4. Single PDF Feasibility Results

The single PDF feasibility experiment evaluated retrieval and answer generation on 10 Open RAGBench questions associated with arXiv:2410.14077v2.

The experiment compared a MiniLM retrieval baseline with BGE-M3 and reranking. It also compared text only generation with generation using selected rendered PDF pages.

Table 8 separates retrieval and generation results because their baseline configurations differ.

Table 8. Retrieval and answer generation performance in the single PDF experiment

Panel A. Retrieval performance		
Metric	MiniLM baseline	BGE-M3 with reranking
Hit@1	0.30	**0.60**
Hit@3	0.50	**0.90**
Hit@5	0.80	**1.00**
Panel B. Answer generation performance		
Metric	Text only	Text with page images
Semantic similarity	0.7079	**0.7310**
Token F1	0.2478	**0.2918**
ROUGE-L	0.2002	**0.2379**

Both generation conditions used the same BGE-M3 evidence. They differed only in whether the selected page images were provided to the model.

Hit@1 increased from 0.30 to 0.60 while Hit@3 increased from 0.50 to 0.90. Hit@5 reached 1.00 under the BGE-M3 retrieval configuration.

Adding selected page images also increased all three generation metrics in this experiment. Semantic similarity increased from 0.7079 to 0.7310. Token F1 increased from 0.2478 to 0.2918 while ROUGE-L increased from 0.2002 to 0.2379.

Full page OCR did not improve retrieval. OCR was therefore retained only as a fallback for pages without extractable text.

These findings support the feasibility of the retrieval and answer generation stages. However, the experiment contained only 10 questions from one document. Its results cannot be generalized to a larger collection of scientific papers.

4.5. Automated PDF Review Evaluation

The automated PDF review pipeline was evaluated through structural checks and manual examination of generated scientific claims.

The manual audit covered four documents: an inverter impedance estimation paper, an LLM fine tuning paper published in *Big Data and Cognitive Computing*, a tabular machine learning benchmark and a Parkinson's disease classification report.

The evaluation examined whether generated extraction records were supported by their cited PDF evidence.

4.5.1. Manual Evidence Support

Evidence support rates varied substantially across documents and generation configurations.

Table 9 reports the proportions of audited extraction records labeled Supported as written. The values are based on records rather than deduplicated scientific propositions.

Table 9. Manual evidence support by document and generation mode

Document	Text retrieval	Hybrid text	Hybrid images
Inverter	20%	27%	32%
Tabular benchmark	∼71%	∼69%	∼75%
Parkinson's disease	76% (28/37)	75% (27/36)	62% (18/29)
BDCC	n = 3	n = 1	n = 11

Percentages represent the proportion of audited extraction records supported as written. BDCC values represent audited record counts because support rates were not computed. The inverter audit is reported descriptively and is not interpreted as the performance of the final pipeline version.

Across the three documents with interpretable support rates, observed values ranged from 20% to 76% across generation modes. Considering only the best mode for each document, the rates ranged from 32% to 76%.

The inverter document had 10 to 19 audited records per mode. The tabular benchmark had 24 to 37 records per mode. BDCC had too few records for a meaningful percentage comparison.

Because rates were calculated over extraction records, repeated claims could contribute more than once. These percentages therefore do not estimate support rates for unique scientific propositions.

The Parkinson's disease report achieved its highest support rate under text retrieval, with 28 of 37 records supported as written.

No fabricated scientific findings were identified among the manually audited structured review outputs. This observation applies only to the audited records and does not establish that the system is free from hallucinations.

4.5.2. Error Analysis

The manual audit identified errors involving evidence retrieval, page attribution, source attribution, numerical extraction and model or metric identification.

Table 10 presents representative examples.

Table 10. Representative errors identified during manual evidence verification

Document	Observed error	Error category
Inverter	Six of seven suspicious values were correct content linked to incorrect pages	Attribution
Inverter	36.7264% was extracted as 3%	Numerical extraction
Tabular benchmark	−0.0195 was extracted as 0.0195	Numerical extraction (sign)
Tabular benchmark	An AUC value was labeled as accuracy	Metric attribution
BDCC	A 219% result from another paper was attributed to the reviewed document	Source attribution
Parkinson's disease	A PatchTST characteristic was attributed to Mamba	Model attribution

These examples demonstrate that finding a numerical value in the source document does not guarantee correct interpretation or attribution. Structural validation alone cannot detect all errors involving numerical signs, metric labels or the scientific context of a reported result.

4.5.3. Page Diversity and Evidence Selection

The inverter paper revealed a limitation in selecting evidence solely according to retrieval rank.

Before applying page diversity, five of the six selected chunks came from page 2. Relevant content from later pages remained outside the selected evidence. One relevant passage appeared at rank 11.

After introducing the page diversity constraint, evidence from pages 4 and 5 entered the selected context.

This case demonstrates how highly ranked overlapping passages can crowd out relevant evidence from other document sections. Page diversity improved evidence coverage in this example.

4.5.4. Structural Reliability and Reviewer Reproducibility

The pipeline applies structural checks to generated extraction and review records. These checks validate evidence references, page associations and numerical fields. They also flag values that cannot be located in the extracted text of the cited page.

Most of these rules came from failures seen in real runs. The model sometimes wrote "page 2" where a bare page number was expected. It put words such as "better" into a numeric field or left a trailing comma inside its JSON. It also cited evidence references that did not exist. Because decoding was greedy, a simple retry usually reproduced the same mistake. Each case was therefore handled by a dedicated normalization or validation rule. These rules are covered by 66 regression tests, each based on a failure observed during development.

One instability turned out to have a technical cause rather than a content related one. Two reviewer runs on the same evidence produced

different recommendations and different gaps. Since decoding was greedy, this pointed to a difference in the input. The prompt showed the model an example evidence reference, taken as the first element returned when iterating over a Python set. By default, Python randomizes string hashing in every new process. The set could therefore be iterated in a different order in each job. The model could then see a slightly different prompt and write a different review.

The example reference was then chosen after sorting the references alphabetically. This made the prompt identical across runs. After this change, two independent runs on the computing cluster with identical evidence produced byte identical review outputs.

This demonstrated that review generation was reproducible across the tested independent runs. It did not establish that the generated scientific assessments were semantically correct.

4.5.5. Semantic Limitations of Structural Validation

The tabular benchmark revealed a limitation that structural validation could not identify. The paper trained a model to predict whether deep learning or tree-based ensemble models perform better on a given dataset. This model achieved an accuracy of 86.1% on a subset of 36 datasets. Deep learning performed better on 11 of these datasets and tree-based ensembles on 25. A classifier that always predicted tree-based ensembles would therefore already achieve 69.4% accuracy.

The pipeline correctly extracted the 86.1% value in seven separate records. However, none of the generated recommendations examined the result in relation to the majority class baseline.

The reported accuracy exceeded this baseline. The limitation was not an incorrect numerical result but the absence of a contextual assessment. This case demonstrates that structural validation can confirm whether a value appears in the cited evidence without establishing whether the generated review adequately evaluates its scientific significance.

4.5.6. Incorrect Absence Claims

The Parkinson's disease report revealed cases in which the reviewer incorrectly identified available information as missing. For example, the reviewer claimed that the dataset size was not specified although page 1 reported 93 Parkinson's disease patients and 73 healthy controls.

Similar statements questioned the availability of validation and preprocessing details despite these being described in the document.

These statements appeared as unresolved checks rather than as recommendations. For the leakage safe cross-validation example, the output explicitly confirmed that the absence claim validation mechanism had moved the statement to an unresolved check because no cited extraction record was labeled as a limitation.

The reviewer therefore sometimes missed information that was available in the paper.

4.5.7. Forced Answer Generation and Abstention

A separate failure was observed during ad hoc question answering on the BDCC document.

When asked about the main dataset, the model was forced to produce a nonempty answer despite insufficient supporting evidence. It generated an unsupported GSM8K record.

When abstention was permitted, the model instead returned an empty answer and described the evidence that had been examined. This observation led to a design change in the ad hoc question answering function ($\mathrm{ask_pdf_question}$), which no longer forces the generation of nonempty answers.

The GSM8K case was part of ad hoc question answering rather than the manually audited structured review outputs. It therefore does not contradict the absence of fabricated scientific findings among those audited records.

This comparison illustrates the importance of allowing abstention when retrieved evidence is insufficient.

Overall, the results demonstrate that evidence grounded scientific review requires more than structurally valid citations. Retrieval quality, numerical interpretation, source attribution and the ability to abstain all influence the reliability of the generated assessments.

5. Discussion

5.1. Contribution of Textual and Visual Evidence

The controlled ablation showed that figure related textual context improved mean semantic similarity compared with unrelated distractor context. The difference was statistically significant but remained below the practical effect threshold. Adding image pixels produced only a small additional improvement that was not statistically significant.

A large part of the average multimodal benefit appears to be carried by the text associated with each figure. Captions and the surrounding discussion may already describe the content of a plot. In such cases, the pixels provide limited additional information. This pattern does not extend to every figure. In the generation size example (Section 4.3.2), the model identified the value of 10^9 only when the figure itself was supplied.

The visual necessity analysis provides a more detailed interpretation. Questions classified as REQUIRED showed a positive mean change when image pixels were available while NOT_NEEDED questions showed a negative mean change.

These findings suggest that the usefulness of image pixels depends on the question and the retrieved evidence. Their limited average contribution may reflect a mixture of questions that require visual interpretation and questions that can be answered from text. However, the visual necessity subset was manually selected and contained only 14 questions. Its results cannot establish how frequently visual evidence is necessary in the full benchmark.

With respect to RQ1, figure related textual context provided a statistically significant but practically small improvement over distractor context. For RQ2, adding image pixels did not produce a statistically significant improvement in the overall evaluation.

5.2. Retrieval Quality and Evidence Attribution

For most questions in the oracle experiment, image retrieval was already complete. This applied to 11 of the 14 questions. Replacing their images with the oracle references left the answers unchanged. Only three questions received different visual inputs. In these cases, semantic similarity improved for two questions and decreased for the third. Selecting the relevant images therefore improved some answers without consistently leading to a better result.

The failed visual evidence example shows that retrieval quality cannot be judged only by whether the correct document or section appears among the retrieved candidates. In this case, the correct section was ranked second after reranking. The selected images, however, were unrelated to the question. Neither the figure context nor the image pixels allowed the model to produce the correct answer.

The PDF review pipeline showed a similar pattern. In the inverter document, highly ranked chunks from a single page initially filled most of the evidence positions. Limiting the number of chunks taken from each page allowed content from other pages to enter the selected evidence. Coverage improved in this case. An overall improvement in review accuracy, however, was not established.

Retrieval configuration also played a role in the single PDF experiment. With BGE-M3 and reranking, Hit@1 increased from 0.30 to 0.60 compared with the MiniLM baseline. Adding rendered pages to the same textual evidence increased all three answer quality metrics. These results are based on 10 questions from a single paper.

The difference between the two multimodal generation experiments is also noteworthy. Adding rendered page images increased mean semantic similarity by approximately 0.023 in the single PDF experiment whereas adding figure pixels increased it by only 0.00123 in the broader benchmark. These results are not directly comparable. The experiments differed in sample size, document scope and visual input representation. The available evidence does not establish which of these differences explains the observed performance gap.

Retrieval coverage, evidence selection and attribution therefore need to be considered separately. In the failed visual evidence example, the relevant section was retrieved. The final evidence was nonetheless

incomplete. A citation may also pass the structural checks while referring to the wrong page or supporting a different statement. In the inverter document, several values were correct but linked to incorrect pages.

With respect to RQ3, the descriptive results suggest that image pixels were more useful for questions that required visual interpretation and when the relevant image was retrieved. However, the small and manually selected subsets do not establish the magnitude of these effects across the benchmark.

5.3. Reliability of Automated Scientific Review

The PDF review evaluation revealed a distinction between structural validity and semantic reliability. The pipeline was designed to enforce valid evidence references, handle malformed structured outputs and retain uncertain statements for verification. These mechanisms made the reports more consistent and easier to audit but did not establish the scientific correctness of their contents.

Several errors in the manual audit could not be resolved through the structural checks. In the tabular benchmark, a value of -0.0195 was extracted without its minus sign. An AUC value from the same paper was labeled as accuracy. In other documents, a result from a cited paper was attributed to the reviewed document. In another case, a characteristic of PatchTST was assigned to Mamba. A value that appears in the PDF therefore does not guarantee that the generated statement interprets it correctly.

In the tabular benchmark, the pipeline extracted the reported accuracy of 86.1% correctly in several records. None of the generated recommendations related this value to the 69.4% majority class baseline. The reported result itself was not incorrect. What the review omitted was an assessment of how the result should be interpreted. Reliable scientific review therefore appears to require contextual reasoning in addition to evidence extraction.

In the Parkinson's disease report, the reviewer treated some available information as missing. For example, it stated that the dataset size was not specified, although page 1 reported 93 patients and 73 healthy controls. A related criticism concerned the leakage safe cross-validation procedure. This statement cited no extraction record labeled as a limitation. It was therefore moved to the unresolved checks instead of remaining a recommendation. As a result, it was not presented as an established weakness of the paper. The underlying interpretation error, however, remained in the output.

In the BDCC document, the model was asked to name the main dataset. When it was required to return a nonempty answer, it produced an unsupported GSM8K record. When abstention was permitted, it returned an empty answer instead. Following this observation, the question answering interface was changed so that it can return no answer when the evidence is insufficient.

Reviewer generation was also affected by an instability in the prompt. The example evidence reference shown to the model could change between runs. After this reference was selected in a fixed order, two independent runs on the computing cluster produced identical review outputs. This confirms reproducibility for the tested configuration. It does not confirm that the generated assessments are semantically correct.

With respect to RQ4, the structural checks linked each finding to a cited page. Together with the reproducible reviewer output, this made it possible to check every audited claim against its source. Semantic errors and unsupported interpretations were nevertheless identified only through manual verification. Examples include the lost minus sign and the missing comparison with the majority class baseline.

5.4. Limitations and Future Work

The results presented above are subject to several limitations.

First, answer quality was primarily evaluated using automatic reference based metrics. The calibration exercise showed that semantic similarity scores overlap across correct, partial and incorrect answers. Consequently, improvements in semantic similarity should not be interpreted as equivalent improvements in factual correctness. The same all-MiniLM-L6-v2 model was used for text retrieval and semantic similarity evaluation. Because retrieval was identical across the controlled conditions, this is unlikely to bias comparisons between conditions. It may still influence the absolute similarity scores.

Second, the visual necessity analysis included only 14 manually selected questions. The subset was not randomly sampled and contained only one HELPFUL question. The oracle experiment changed the actual image inputs for just three questions. These subsets show how image pixels and image retrieval can influence individual answers. They cannot show how often this occurs across the full benchmark.

Third, the single PDF feasibility experiment covered one document and 10 questions. Its retrieval and generation improvements provide an initial demonstration rather than evidence of consistent performance across document types, layouts or scientific disciplines.

Fourth, the manual PDF review audit covered four documents with different numbers of extraction records. The BDCC document, for example, produced between 1 and 11 records per generation mode. This was too few for a meaningful support rate. In the other documents, some records repeated the same underlying claim. All records were checked by a single annotator. The resulting percentages are therefore descriptive. They should not be read as representative rates for the scientific literature or as estimates of agreement between independent reviewers.

Fifth, the paired t-test used for the multimodal ablation was chosen after the results had been inspected. The original protocol did not specify a test in advance. Holm correction was applied. The analysis was nevertheless not fully prespecified. The power calculation relied on a standard deviation of 0.10222, estimated from the observed paired differences. It also relied on the chosen practical threshold of 0.02. The reported significance and sensitivity should be interpreted with these choices in mind.

Sixth, the controlled ablation used different GPU architectures across generation conditions. T and T+D were generated on a V100 while T+C and T+C+P were generated on an A100. Although all conditions used greedy decoding, small hardware dependent numerical differences could affect generated answers. Repeatability was verified within the V100 configuration for T and T+D but cross hardware equivalence was not established. This limitation primarily concerns the T+C versus T+D comparison. The pixel comparison between T+C+P and T+C was conducted on the same A100 and was not subject to this cross hardware difference.

Seventh, the controlled ablation used a single generator model and a fixed retrieval configuration. The observed results may differ with other generators, retrievers, evidence selection strategies or visual input formats. In particular, the limited average contribution of image pixels should not be generalized to all multimodal RAG systems.

Eighth, the automated PDF pipeline used Qwen2.5-VL for both evidence extraction and reviewer generation. These stages therefore did not provide independent model based verification. An incorrect extraction could be accepted or repeated by the reviewer, leaving systematic interpretation errors undetected by the structural validation checks.

Finally, the PDF pipeline was evaluated primarily on documents with extractable text. Full page OCR did not improve retrieval in the feasibility experiment and scanned PDFs were not established as a supported document category. The system should not be assumed to generalize to image only or heavily degraded PDFs.

Several of these limitations point to concrete next steps. The visual necessity findings rest on 14 manually selected questions. A larger and independently sampled subset would show how often image pixels are actually needed. The failed retrieval example suggests that figure retrieval could benefit from methods that align each figure more closely with the questions it answers. For the PDF review pipeline, evaluation on a broader collection of papers would be required. Human evaluation could then assess claim support, numerical interpretation and the usefulness of recommendations as separate criteria. Since extraction and review currently rely on the same model, an independent verification model may help detect systematic errors. Controlled runs on both GPU

types could clarify whether hardware affects the generated answers. Together, these steps would help separate retrieval limitations from errors in generation and scientific reasoning. Structural validity would still not replace semantic verification.

6. Conclusion

This project investigated the contribution of textual and visual evidence to scientific question answering and developed an automated pipeline for evidence grounded scientific paper review.

The controlled experiment compared four evidence configurations on 982 image associated questions from Open RAGBench. Text describing the figures raised mean semantic similarity by 0.01243 relative to distractor context. This difference was statistically significant. It remained, however, below the practical threshold of 0.02. Supplying the figure pixels added only 0.00123, a difference that was not statistically significant. In a descriptive analysis of 14 questions, the images appeared more useful when the question depended on the figure itself and the relevant image had been retrieved.

The second part of the project produced a pipeline for reviewing individual scientific PDFs. Its main lesson was that correct citations did not ensure correct content. Retrieval with BGE-M3 and a reranker raised Hit@1 on a single paper from 0.30 to 0.60 compared with MiniLM. A vision language model generated the findings. Each citation was checked against the retrieved pages. The manual audit of four documents nevertheless revealed errors. Some values were attributed to the wrong page or source. Others were interpreted incorrectly. In some cases, information present in the paper was reported as missing.

Overall, the results indicate that figure related textual context captures much of the average benefit observed in the evaluated multimodal RAG configuration while image pixels may provide additional value for visually dependent questions. They also highlight the importance of reliable evidence selection and semantic verification in automated scientific review. The developed pipeline provides a foundation for further investigation of evidence grounded scientific analysis while retaining human verification for claims whose correctness cannot be established through structural checks alone.

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